

Utkarsh Tyagi

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EDUCATION

University of Maryland, College Park

College Park, Maryland

M.S. in Computer Science, Advised by Prof. Dinesh Manocha - GPA 3.9/4.0

08/2023 – 05/2025

Delhi Technological University (Formerly Delhi College of Engineering)

Delhi, India

B.Tech in Computer Science - GPA 8.7/10, Major GPA 9.05/10

2017 – 2021

SELECTED PUBLICATIONS

MULTIMODAL LEARNING

- Audio MultiChallenge: A Multi-Turn Evaluation of Spoken Dialogue Systems on Natural Human Interaction
ACL 2026 (Under Review)
- Beyond Seeing: Evaluating Multimodal LLMs on Tool-Enabled Image Perception, Transformation, and Reasoning
ICLR 2026 (Under Review)
- MMAU: A Massive Multi-Task Audio Understanding and Reasoning Benchmark
ICLR 2025 (Spotlight)
- VDGD: Mitigating LLM Hallucinations in Cognitive Prompts by Bridging the Visual Perception Gap
ICLR 2025
- ProSE: Diffusion Priors for Speech Enhancement
NAACL 2025
- CompA: Addressing the Gap in Compositional Reasoning in Audio-Language Models
ICLR 2024
- GAMA: A Large Audio-Language Model with Advanced Audio Understanding and Complex Reasoning Abilities
EMNLP 2024
- LipGER: Visually-Conditioned Generative Error Correction for Robust Automatic Speech Recognition
InterSpeech 2024
- AdVerb: Visually Guided Audio Dereverberation
ICCV 2023
- MMER: Multimodal Multi-task Learning for Speech Emotion Recognition
InterSpeech 2023

NATURAL LANGUAGE PROCESSING (NLP)

- ABEX: Data Augmentation for Low-Resource NLU via Expanding Abstract Descriptions
ACL 2024
- ACLM: Selective-Denoising based Data Augmentation for Low-Resource Complex NER
ACL 2023
- DALE: Generative Data Augmentation for Low-Resource Legal NLP
EMNLP 2023
- BioAug: Conditional Generation based Data Augmentation for Low-Resource Biomedical NER
SIGIR 2023

INDUSTRY EXPERIENCE

Scale AI

San Francisco, California

Research Engineer - Audio/Multimodal

June 2025 – Present

- Led the post-training data collection for speech to speech (S2S) assistants and ran large scale experiments using audio rubrics to improve instruction following in multi-turn spoken dialogues
- Researching rubrics-as-rewards for multimodal LLMs targeting perception abilities with expert-graded supervision
- Designed the AI Research Scientist benchmark and leaderboard that evaluates LLMs ability to predict the outcomes of real-world scientific experiments (**Under review ICML 2026**)
- Partnering with frontier labs to diagnose next-gen model loss/failure regions (audio & multimodal) and propose targeted data programs (task design, rubric spec, and evaluation) to close capability gaps

Samsung SARC/ACL

Research Intern - Efficient GenAI Inference

San Jose, California

May 2024 – August 2024

- Optimized state-of-the-art Large Language Models for deployment on resource-constrained devices, improving latency and memory footprint.
- Fine-tuned LLMs and developed tools with advanced Retrieval-Augmented Generation techniques, leveraging internal documentation to enhance knowledge retrieval and response accuracy.

Atlassian India LLP

Software Development Engineer 2

Bangalore, Karnataka

July 2021 – August 2023

- Worked on the JIRA Issue Create Experience to deliver a reliable and performant interface, which helped improve key business metrics and customer satisfaction ratings.
- Achieved **99.996% frontend reliability** and scaled the experience for over 250k customers with more than 2M issues created every day. Improved performance TTI from 3.9s to **1.2s** using modern technology stack.

Software Engineer Intern

Jan 2021 – June 2021

- Identified limitations and bottlenecks in the existing legacy API implementation and proposed comprehensive improvements for system revamping. Utilized this performant API to develop the user interface.
- Reduced the **latency** of modal load by **95.57%** by avoiding redundant server-side database queries and scaled to 1M customers.

Samsung R&D Institute Bangalore

Research Intern - Speech AI

Bangalore, Karnataka

May 2020 – July 2020

- Worked with the **Voice Intelligence** team of Samsung to improve Bixby's wakeup word detection.
- Researched keyword spotting techniques and built an efficient on-device LSTM-CTC based, vocabulary-independent keyword spotter

RESEARCH EXPERIENCE

GAMMA Lab, University of Maryland

Research Assistant

College Park, Maryland

August 2022 – Present

- Researched multimodal learning; working on video representation learning for long-form video understanding
- Also focused on low-resource (labeled data and compute) learning with speech, NLP, or vision applications. In this area, I solved problems using self-supervised learning, synthetic data augmentation, etc.
- Published at **ICLR**, **ICCV**, **ACL**, **EMNLP**, **InterSpeech**, **SIGIR**, **NAACL**
- Advised by [Prof. Dinesh Manocha](#)

Multimodal Digital Media Analysis Lab

Machine Learning Researcher

IIITD, Delhi

April 2022 – March 2023

- Worked under the supervision of [Dr. Rajiv Ratn Shah](#) in the areas of Speech and Language Processing.
- Worked on multi-lingual automatic speech scoring systems for low-resource Indian languages.
- Also explored novel architectures to detect implicit hate speech in online conversations. Paper accepted at **AAAI 2023 Defactify**

PATENTS

- Apparatuses, methods, and computer program products for generating an abstract context summary scheduling interface configured for scheduling and outputting abstract context summaries for multi-party communication channels
U.S. Patent Application No. 17/936,695 (Patent Pending)
- Apparatuses, methods, and computer program products for generating and selectively outputting abstract context summaries for multi-party communication channels
U.S. Patent Application No. 17/936,705 (Patent Pending)

TECHNICAL SKILLS

Languages: (Highly Proficient) Python, JavaScript, C++ (Moderate) Java, SQL

Frameworks: Pytorch, FastAI, Tensorflow, Scikit-Learn, ReactJS, Jest, Enzyme, NLTK

ACHIEVEMENTS

- Awarded **\$7,000 Fellowship** at UMD: Project on EEG-based Brain-Computer Interaction using LLMs
- **Reviewer** for ICLR 2025, ACL ARR 2024 (Feb, April, June, October), ICASSP 2025, EMNLP 2024
- **People's choice award 2021 & 2022:** Atlassian's internal hackathon
- **Winner** of Digital Management Inc. India Hackathon, 2019
- **2nd position** in Student Hackday 2019 organised by Skillenza
- **Barclays India Hackathon 2019 Top 4**, All India
- Achieved **5 star rating** on **CodeChef** platform for competitive programming
- **JEE Mains 2017: 99.49%** percentile, Pan India